

1. How to print all the arguments provided to the script?

```
sneha_mhasalkar@cloudshell:~/day4$ ./que1.sh hi this is java
hi
this
is
java
sneha_mhasalkar@cloudshell:~/day4$ █
```

2. Print a given number, in reverse order using a Shell script such that the input is provided using command Line Argument only.

To solve this problem i have use while loop and have take command line arguments

```
sneha_mhasalkar@cloudshell:~/day4$ cat que2.sh
#!/bin/bash
#
num=$1
rev=0
while [ $num -gt 0 ]
do
    d=`expr $num % 10`
    rev=`expr $rev \* 10 + $d`
    num=`expr $num / 10`
done
echo "reverse number : $rev "
sneha_mhasalkar@cloudshell:~/day4$ █
```

```
sneha_mhasalkar@cloudshell:~/day4$ vi que2.sh
sneha_mhasalkar@cloudshell:~/day4$ ./que2.sh 123
reverse number : 321
sneha_mhasalkar@cloudshell:~/day4$ █
```

3. Write a script to print the first 10 elements of Fibonacci series.

```
#!/bin/bash
#
a=0
b=1
n=$1

for ((i=1;i<=$n;i++))
do
    echo -n " $a"
    c=$((a + b))
    a=$b
    b=$c
done
echo
```

```
sneha_mhasalkar@cloudshell:~/day4$ ./que3.sh 5
0 1 1 2 3
sneha_mhasalkar@cloudshell:~/day4$ █
```

4. How to add two strings?

```
sneha_mhasalkar@cloudshell:~/day4$ ./que4.sh
add 2 string : abcbcc
sneha_mhasalkar@cloudshell:~/day4$ █
```

5. Write a shell script using Logical AND operator to check for the given

Expected Output1:

enter value for a

0

enter value for b

2

enter value for c

10

as Expected the values are 0 2 10

Expected Output2: (Note these values can be anything)

enter value for a

5

enter value for b

3

enter value for c

4

not as expected

```
sneha_mhasalkar@cloudshell:~/day4$ ./que5.sh 0 2 10
excepted values are 0 2 10
sneha_mhasalkar@cloudshell:~/day4$ ./que5.sh 0 2 5
Not as excepted
sneha_mhasalkar@cloudshell:~/day4$ █
```

6. Write a shell script on time based input meaning input has to be read within 5 sec or else it will show time out (time out and return failure if a complete line of input is not read within TIMEOUT seconds)

Expected Output1:

enter a string

TIME OUT

```
sneha_mhasalkar@cloudshell:~/day4$ ./que6.sh
enter the input
time out
sneha_mhasalkar@cloudshell:~/day4$ █
```

Expected Output 2:

enter a string

farha

given string=farha

```
sneha_mhasalkar@cloudshell:~/day4$ ./que6.sh
enter the input
hello
given string = hello
sneha_mhasalkar@cloudshell:~/day4$ █
```

7. Write a shell script to check whether the string entered by the user is null or not. if not then print the string and its length

Expected Output1:

enter a string

No input

```
sneha_mhasalkar@cloudshell:~/day4$ ./que7.sh
string empty
sneha_mhasalkar@cloudshell:~/day4$
```

Expected Output 2:

enter a string

farhaqureshi

given string =farhaqureshi

length= 12

```
sneha_mhasalkar@cloudshell:~/day4$ ./que7.sh hello
given string: hello
length = 5
sneha_mhasalkar@cloudshell:~/day4$
```

8. Write a shell script to print sum of squares till 5 i.e $1^2+2^2+3^2+4^2+5^2$ using for loop

Expected Output:

sum of squares =55

```
sneha_mhasalkar@cloudshell:~/day4$ cat que8.sh
#!/bin/bash
#
n=$1

for (( i=1; i<=n; i++ ))
do
    d=$(( i*i ))
    sum=$(( sum+d ))
done

echo "sum of the number : $sum"

sneha_mhasalkar@cloudshell:~/day4$
```

```
sneha_mhasalkar@cloudshell:~/day4$ ./que8.sh 5
sum of the number : 55
sneha_mhasalkar@cloudshell:~/day4$ █
```

9. Write a shell script to check whether is given char is alpha or digit or other

Expected Output1:

enter a charA

An Alphabet

Expected Output2:

enter a char#

other char

Expected Output3:

enter a char7

Digit

```
sneha_mhasalkar@cloudshell:~/day4$ ./que9.sh A
it is character
sneha_mhasalkar@cloudshell:~/day4$ ./que9.sh 7
it is digit
sneha_mhasalkar@cloudshell:~/day4$ ./que9.sh #
it is other special symbol
sneha_mhasalkar@cloudshell:~/day4$ █
```

10. Write a shell script to display menu like “1. Date, 2. Cal, 3. Ls, 4. Pwd, 5. Exit” and execute the commands depending on user choice.

For this question i use the switch case method

```
sneha_mhasalkar@cloudshell:~/day4$ ./que10.sh
1.Date
2.Cal
3.Ls
4.pwd
5.exit
enter the user choice
1
Mon Jun 22 08:16:28 AM UTC 2026
sneha_mhasalkar@cloudshell:~/day4$ █
```

11. Write a shell script to accept the name from the user and check whether user entered name is file or directory. If name is file display its size and if it is directory display its contents (list of files in the directory).

```
this is the directory
de/  demo2/  demo3/  demo4.txt
```

```
sneha_mhasalkar@cloudshell:~/day2$ ./q11.sh file.txt
this is file
Size: 0 (Total details: -rw-rw-r-- 1 sneha_mhasalkar sneha_mhasalkar 0 Jun
19 11:40 file.txt)
sneha_mhasalkar@cloudshell:~/day2$ █
```

12. Write a Program to find the greatest of three numbers

```
bash: ./ques11.sh: No such file or directory
sneha_mhasalkar@cloudshell:~/day2$ ./ques11.sh 10 8 2
a is greater than both
sneha_mhasalkar@cloudshell:~/day2$ ./ques11.sh 10 8 20
c is greater
sneha_mhasalkar@cloudshell:~/day2$ █
```

13. Write a program to calculate gross salary if the DA is 40%, HRA is 20% of basic salary. Accept basic salary from user and display gross salary (Result can be floating point value).

```
sneha_mhasalkar@cloudshell:~/day4$ vi ques13.sh
sneha_mhasalkar@cloudshell:~/day4$ ./ques13.sh
enter the basic sal
20000
DA: 8000
HRA: 4000
Gross Salary= 32000
sneha_mhasalkar@cloudshell:~/day4$ █
```

14. Write a shell script that computes the gross salary of a employee according to the following rules:

- a. . If basic salary is < 1500 then HRA =10% of the basic and DA =90% of the basic.**
- b. If basic salary is >=1500 then HRA =Rs500 and DA=98% of the basic**

The basic salary is entered interactively through the keyboard.

```
sneha_mhasalkar@cloudshell:~/day4$ ./que14.sh
enter the basic salary
1000
gross salary: 2000
DA : 900
HRA : 100
sneha_mhasalkar@cloudshell:~/day4$ ./que14.sh
enter the basic salary
5200
gross salary: 10796
DA : 5096
HRA : 500
sneha_mhasalkar@cloudshell:~/day4$ █
```